

QUALITY ASSURANCE PLAN

Approved Technical & Commercial Configuration Summary

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SALES REQUEST SUMMARY

ID

9176C631-32BB-4924-A073-0D34DCB40D95

PROJECTCODE

SR-CUS-P2-M10-20261209-0001

CUSTOMERNAME

1

MODULEMANUFACTURINGPLANT

P2

CELLTYPE

DCR

MODULECELLTYPE

M10

CELLTECH

TOPCon

CUTCELLS

72

CERTIFICATIONREQUIRED

BIS + IEC + 3xIEC

WATTAGEBINNINGDIST

Bin	Value
540	33.3
550	33.3
560	33.4

RFQORDERQTYMW

100

PREMIERBIDDEDORDERQTYMW

100

DELIVERYSTARTDATE

2026-12-09

DELIVERYENDDATE

2027-03-09

PROJECTLOCATION

hyderabad

CABLELENGTHREQUIRED

1,000

QAATYPE
Premier Energies
QAATYPEATTACHMENTURL
-
BOMFROM
Premier Energies
PRIMARYBOMATTACHMENTURL
-
INLINEINSPECTION
no
PDI
no
CELLPROCUREDBY
Customer
AGREEDCTM
1,000
FACTORYAUDITTENTATIVEDATE
2026-01-10
XPITCHMM
1,400
TRACKERDETAILS
790
PRIORITY
low
REMARKS
test
OTHERATTACHMENTS
-
CREATEDBY
sales
CREATEDAT
2025-12-09T04:05:07.678Z
BOM

See BOM Summary section.

MQP SPECIFICATIONS

Parameter	Customer Spec	Premier Spec	Remarks / Notes	Match	Last Edit
IQC	As per approved BOM	As per approved BOM	-	YES	-
IQC	-	Tolerance: ±1.5mm	-	NO	production
IQC	-	Tolerance: ±1.5mm	-		-
IQC	-	3.2mm±0.2mm, 4.0mm±0.3mm, 2.0mm±0.2mm	-		-

Parameter	Customer Spec	Premier Spec	Remarks / Notes	Match	Last Edit
IQC	-	Normal $\geq 91.5\%$ & AR Coated $\geq 93.5\%$	-	☐	- -
IQC	-	Prismatic/Matt, Matt/Matt	-	☐	- -
IQC	-	$\leq 3\text{mm}$	-	☐	- -
IQC	-	Size 0 - 1mm, Qty: 5, Size 0 -2mm Qty: 3, Size >2mm not allowed	-	☐	- -
IQC	-	L: <5-10 mm,W<1mm Qty 2 nos	-	☐	- -
IQC	-	L < 10 W <1mm, Qty: 1	-	☐	- -
IQC	-	No dirt, finger prints, edge break,roller print & bright line, rainbow	-	☐	- -
IQC	-	$\leq 150\text{ PPM}$	-	☐	- -
IQC	-	Broken Pieces should be ≥ 40 nos per 50 x 50 mm area	-	☐	- -
IQC	-	No Breakage	-	☐	- -
IQC	-	As per approved BOM	-	☐	- -
IQC	-	As per manufacturer criteria	-	☐	- -
IQC	-	Tolerance: $\pm 2\text{ mm}$,	-	☐	- -
IQC	-	Shelf Life 6/ 9 months	-	☐	- -
IQC	-	75% - 95%	-	☐	- -
IQC	-	$\geq 70\text{N/cm}$ $\geq 60\text{N/cm}$	-	☐	- -
IQC	-	0	-	☐	- -
IQC	-	$\leq 360\text{nm}$	-	☐	- -
IQC	-	$1 \times 10^{14}\text{ Ohm.cmm}$	-	☐	- -
IQC	-	TD < 1.5% , MD < 3.0 %	-	☐	- -
IQC	-	No dust and other foreign material; no breakage,no stain.	-	☐	- -
IQC	-	As per approved BOM	-	☐	- -

Parameter	Customer Spec	Premier Spec	Remarks / Notes	Match	Last Edit
IQC	-	Tolerance : $\pm 10\%$	-	☐	- -
IQC	-	Tolerance: $\pm 2\%$	-	☐	- -
IQC	-	Tolerance: $\pm 2\%$	-	☐	- -
IQC	-	12/ 24 months	-	☐	- -
IQC	-	$\leq 5\text{N/Cm}$	-	☐	- -
IQC	-	23 kV	-	☐	- -
IQC	-	$\leq 2.0 \text{ g}/(\text{m} \cdot \text{day})$	-	☐	- -
IQC	-	1500 V	-	☐	- -
IQC	-	$\geq 60\text{N/cm}$	-	☐	- -
IQC	-	TD < 0.5% , MD < 1.5 %	-	☐	- -
IQC	-	No wrinkle, dirty, bump, color difference.	-	☐	- -
IQC	-	As per approved BOM	-	☐	- -
IQC	-	As per approved BOM data sheet	-	☐	- -
IQC	-	Tolerance: $\pm 0.25\text{mm}$	-	☐	- -
IQC	-	Tolerance: $\pm 0.25\text{mm}$	-	☐	- -
IQC	-	As per Datasheet	-	☐	- -
IQC	-	-	-	☐	- -
IQC	-	Front side: ≥ 7 of the 14 solder joints on the same busbar force of $\geq 0.9\text{N}$, and it is not allowed to exceed two consecutive solder joints < 0.9N ≥ 2 of the 4 solder joints on the same busbar force of $\geq 0.8\text{N}$, and the whole piece is not allowed to exceed two consecutive solder joints below 0.8N when cut in half Backside Judgment Criteria: Each peak $\geq 1.25\text{N}$ There is ≥ 1 tensile force peak value $\geq 0.8\text{N}$ in 2 segments on the same busbar, and two consecutive welding points are not allowed to be < 0.8N when the whole piece is cut in half, and the average effective welding value is $\geq 0.5\text{N}$	-	☐	- -
IQC	-	$\pm 0.1\%$ nominal power	-	☐	- -
IQC	-	No visible cells crack, cells broken, any extended V shape notch are not allowed	-	☐	- -

Parameter	Customer Spec	Premier Spec	Remarks / Notes	Match	Last Edit
IQC	-	As per approved BOM	-	☒	-
IQC	-	As per approved BOM	-	☒	-
IQC	-	As per approved BOM	-	☒	-
IQC	-	As per approved BOM	-	☒	-
IQC	-	Sn/Pb=60%/40%	-	☒	-
IQC	-	≤0.0225 Ohm.mm2 / One Meter	-	☒	-
IQC	-	Front side: ≥7 of the 14 solder joints on the same busbar force of ≥0.9N, and it is not allowed to exceed two consecutive solder joints < 0.9N ≥2 of the 4 solder joints on the same busbar force of ≥0.8N, and the whole piece is not allowed to exceed two consecutive solder joints below 0.8N when cut in half Backside Judgment Criteria: Each peak ≥ 1.25N There is ≥1 tensile force peak value ≥0.8N in 2 segments on the same busbar, and two consecutive welding points are not allowed to be < 0.8N when the whole piece is cut in half, and the average effective welding value is ≥0.5N	-	☒	-
IQC	-	As per COC	-	☒	-
IQC	-	≤5mm/mtr	-	☒	-
IQC	-	Flatness,no stain,no deformation,no crystal,no coating-missing.	-	☒	-
IQC	-	As per approved BOM	-	☒	-
IQC	-	As per approved BOM	-	☒	-
IQC	-	As per approved BOM	-	☒	-
IQC	-	Sn/Pb=60%/40%	-	☒	-
IQC	-	≥3N/mm	-	☒	-
IQC	-	As per COC	-	☒	-
IQC	-	≤5mm/mtr	-	☒	-
IQC	-	Flatness,no stain,no deformation,no crystal,no coating-missing	-	☒	-
IQC	-	As per approved BOM	-	☒	-
IQC	-	As per approved drawing	-	☒	-

Parameter	Customer Spec	Premier Spec	Remarks / Notes	Match	Last Edit
IQC	-	As per approved drawing	-	☐	- -
IQC	-	As per approved drawing	-	☐	- -
IQC	-	≥15 μm	-	☐	- -
IQC	-	As per approved drawing	-	☐	- -
IQC	-	As per approved drawing	-	☐	- -
IQC	-	As per approved drawing	-	☐	- -
IQC	-	As per approved drawing	-	☐	- -
IQC	-	Embossed	-	☐	- -
IQC	-	Same colour, smoot hand flat,no water stain, oil stain, contamination, scratch.coating film no broken	-	☐	- -
IQC	-	≥12 HW	-	☐	- -
IQC	-	As per approved BOM	-	☐	- -
IQC	-	Actual length ±5mm	-	☐	- -
IQC	-	According to Drawings	-	☐	- -
IQC	-	As per Spec	-	☐	- -
IQC	-	As per Spec	-	☐	- -
IQC	-	According to Drawings, No crack	-	☐	- -
IQC	-	As per approved Rating	-	☐	- -
IQC	-	As per approved Rating and embossed on JB	-	☐	- -
IQC	-	1.Diode polarity, quantity is right, terminal should be complete. 2. No damage.	-	☐	- -
IQC	-	>80N	-	☐	- -
IQC	-	As per approved BOM	-	☐	- -
IQC	-	As per supplier MFD	-	☐	- -

Parameter	Customer Spec	Premier Spec	Remarks / Notes	Match	Last Edit
IQC	-	12 months	-	☐	- -
IQC	-	>25Kg for 24 hr	-	☐	- -
IQC	-	As per Supplier Spec	-	☐	- -
IQC	-	As per Supplier Spec	-	☐	- -
IQC	-	≥17KV/mm	-	☐	- -
IQC	-	No bubble, smooth,thick liquid,no damage package, Color	-	☐	- -
IQC	-	As per approved BOM	-	☐	- -
IQC	-	As per supplier MFD	-	☐	- -
IQC	-	12 months	-	☐	- -
IQC	-	0.801-0.810	-	☐	- -
IQC	-	Transparent, colorless	-	☐	- -
IQC	-	2% - 5%	-	☐	- -
IQC	-	As per approved BOM	-	☐	- -
IQC	-	As per supplier MFD	-	☐	- -
IQC	-	6/9/12 months	-	☐	- -
IQC	-	Should be free of dirt, foreign material, Black Spot, no tear & wear, Color	-	☐	- -
IQC	-	≥17KV/mm	-	☐	- -
IQC	-	White	-	☐	- -
IQC	-	As per approved BOM	-	☐	- -
IQC	-	As per Supplier spec	-	☐	- -
IQC	-	Read/ Write	-	☐	- -
IQC	-	No flaw,mildew,dirt,delamination, hole. Obvious defect by naked eye not allowed.	-	☐	- -

Parameter	Customer Spec	Premier Spec	Remarks / Notes	Match	Last Edit
IQC	-	As per specification	-	☐	- -
IQC	-	No damage	-	☐	- -
IQC	-	As per specification	-	☐	- -
IQC	-	No damage	-	☐	- -
IQC	-	As per specification/ Drawing	-	☐	- -
IQC	-	No damage, moisture free, fungus free, Termite free, fitting properly with nails.	-	☐	- -
IQC	-	As per specification	-	☐	- -
IQC	-	As per specification	-	☐	- -
IQC	-	No damage	-	☐	- -
IQC	-	As per specification	-	☐	- -
IQC	-	As per approved drawing	-	☐	- -
IQC	-	No damage, moisture free	-	☐	- -
IQC	-	As per specification	-	☐	- -
IQC	-	No damage	-	☐	- -
IQC	-	As per specification	-	☐	- -
IQC	-	No damage	-	☐	- -
IPQC	-	Temp: 25±2°C, Humidity: ≤40-70%	-	☐	- -
IPQC	-	As per approved BOM	-	☐	- -
IPQC	-	As per Job card	-	☐	- -
IPQC	-	No crack, same color, Same Wp, Same Efficiency.	-	☐	- -
IPQC	-	As per Job card	-	☐	- -
IPQC	-	Should be as per required size in given Job card, no zigzag cutting	-	☐	- -

Parameter	Customer Spec	Premier Spec	Remarks / Notes	Match	Last Edit
IPQC	-	Refer production order	-	☐	- -
IPQC	-	According to the BOM of production plan Supplier and Type are right	-	☐	- -
IPQC	-	6 or 9 months or as per manufacturer datasheet	-	☐	- -
IPQC	-	Not allowed	-	☐	- -
IPQC	-	Tolerance : $\pm 10\%$	-	☐	- -
IPQC	-	EVA: +2mm	-	☐	- -
IPQC	-	According to the BOM of production plan. Supplier and Type are right	-	☐	- -
IPQC	-	According to the BOM of production plan. Supplier and Type are right	-	☐	- -
IPQC	-	According to the BOM of production plan. Supplier and Type are right	-	☐	- -
IPQC	-	According to the BOM of production plan. Supplier and Type are right	-	☐	- -
IPQC	-	Not allowed for M/P	-	☐	- -
IPQC	-	According to the BOM of production plan. Supplier and Type are right	-	☐	- -
IPQC	-	According to the BOM of production plan. Supplier and Type are right	-	☐	- -
IPQC	-	According to the BOM of production plan. Supplier and Type are right	-	☐	- -
IPQC	-	According to the BOM of production plan. Supplier and Type are right	-	☐	- -
IPQC	-	According to the BOM of production plan. Supplier and Type are right	-	☐	- -
IPQC	-	According to the BOM of production plan. Supplier and Type are right	-	☐	- -
IPQC	-	As per specification/Drawing	-	☐	- -
IPQC	-	As per specification/Drawing	-	☐	- -
IPQC	-	Temperature 230 $\pm$ 20 degrees	-	☐	- -
IPQC	-	Front side: ≥ 5 of the 7 solder joints on the same busbar have a peak tensile force of $\geq 0.9\text{N}$, and it is not allowed to exceed two consecutive solder joints $< 0.9\text{N}$ Backside : There is ≥ 4 of the 6 solder joints on the same buusbar peak value $\geq 0.9\text{N}$ and teo consecutive welding ponits are not allowed to be less than 0.9N. There is: ≥ 3 of the solder joints on the busbar peak value $\geq 0.9\text{N}$.	-	☐	- -

Parameter	Customer Spec	Premier Spec	Remarks / Notes	Match	Last Edit
IPQC	-	According to the BOM of production plan Supplier and size is right	-	☐	- -
IPQC	-	According to the BOM of production plan Supplier and size is right	-	☐	- -
IPQC	-	Flatness,no stain,no deformation,no crystal,no coating-missing。	-	☐	- -
IPQC	-	≥3N/mm	-	☐	- -
IPQC	-	As per required current	-	☐	- -
IPQC	-	As per specification/Drawing	-	☐	- -
IPQC	-	Not allowed	-	☐	- -
IPQC	-	As per specification/Drawing	-	☐	- -
IPQC	-	As per specification/Drawing	-	☐	- -
IPQC	-	As per specification/Drawing	-	☐	- -
IPQC	-	As per specification/Drawing	-	☐	- -
IPQC	-	As per approved BOM	-	☐	- -
IPQC	-	As per approved BOM	-	☐	- -
IPQC	-	According to given position in Machine	-	☐	- -
IPQC	-	According to the BOM of production plan Supplier and size is right	-	☐	- -
IPQC	-	As per specification/Drawing	-	☐	- -
IPQC	-	According to the BOM of production plan Supplier and size is right	-	☐	- -
IPQC	-	As per specification/Drawing	-	☐	- -
IPQC	-	As per agreed EL / Visual criteria	-	☐	- -
IPQC	-	As per agreed EL / Visual criteria	-	☐	- -
IPQC	-	As per agreed EL / Visual criteria	-	☐	- -
IPQC	-	As per agreed EL / Visual criteria	-	☐	- -

Parameter	Customer Spec	Premier Spec	Remarks / Notes	Match	Last Edit
IPQC	-	As per agreed EL / Visual criteria	-	☐	- -
IPQC	-	As per agreed EL / Visual criteria	-	☐	- -
IPQC	-	As per agreed EL / Visual criteria	-	☐	- -
IPQC	-	According to Gel Content Test/ Adhesion Test	-	☐	- -
IPQC	-	According to Gel Content Test/ Adhesion Test	-	☐	- -
IPQC	-	75 % - 95 %	-	☐	- -
IPQC	-	Glass to EVA $\geq$ 70N/cm Glass to Back sheet $\geq$ 60N/cm	-	☐	- -
IPQC	-	± 2 % of the set point	-	☐	- -
IPQC	-	Smoothness (No bark's, Lump's, Burr's etc)	-	☐	- -
IPQC	-	≥ 700 Lux	-	☐	- -
IPQC	-	As per visual inspection criteria	-	☐	- -
IPQC	-	Dimension Tolerance ± 1.5 mm	-	☐	- -
IPQC	-	± 1 mm	-	☐	- -
IPQC	-	± 1 mm	-	☐	- -
IPQC	-	≤ 0.5 mm	-	☐	- -
IPQC	-	≤ 0.5 mm	-	☐	- -
IPQC	-	Diagonal Differences ≤ 3 mm	-	☐	- -
IPQC	-	As per approved BOM	-	☐	- -
IPQC	-	6 or 9 or 12 Months / As per manufacturer data sheet	-	☐	- -
IPQC	-	Should be uniform	-	☐	- -
IPQC	-	45 $\pm$ 5 gm/mtr	-	☐	- -
IPQC	-	As per approved BOM	-	☐	- -

Parameter	Customer Spec	Premier Spec	Remarks / Notes	Match	Last Edit
IPQC	-	As per approved BOM	-	☐	-
IPQC	-	6 or 9 or 12 Months / As per manufacturer data sheet	-	☐	-
IPQC	-	No overflow or ZigZag	-	☐	-
IPQC	-	10±2 gm/pc	-	☐	-
IPQC	-	No overflow	-	☐	-
IPQC	-	Difference between In time and out time should be ≥4 hr , Temperature 25±5°C, Humidity ≥50 %	-	☐	-
IPQC	-	Difference between In time and out time should be ≥4 hr , Temperature 25±5°C, Humidity ≥50 %	-	☐	-
IPQC	-	Difference between In time and out time should be ≥4 hr , Temperature 25±5°C, Humidity ≥50 %	-	☐	-
IPQC	-	Difference between In time and out time should be ≥4 hr , Temperature 25±5°C, Humidity ≥50 %	-	☐	-
IPQC	-	Should be smooth, no burr	-	☐	-
IPQC	-	No dirt, melted EVA on module	-	☐	-
IPQC	-	Excess sealant need to be removed	-	☐	-
IPQC	-	No dirt, melted, Smear EVA on glass	-	☐	-
IPQC	-	As per technical datasheet	-	☐	-
IPQC	-	As per technical datasheet	-	☐	-
IPQC	-	As BIS standard	-	☐	-
IPQC	-	(The measured insulation resistance times the area of module shall not be <40 Mohms/mtr ² @ system voltage) 1.5KV for 3 sec, ≥40 Mohms*mtr ²	-	☐	-
IPQC	-	Test conditions: (System voltage X 2 + 1000) X 120%, Current Leakage is ≤ 50 μA	-	☐	-
IPQC	-	2.5 x of fuse rating for 3 sec, ≤0.1Ω	-	☐	-
IPQC	-	As per agreed standard	-	☐	-
IPQC	-	1.5KV for 120 sec, ≥40 Mohms*mtr ²	-	☐	-

Parameter	Customer Spec	Premier Spec	Remarks / Notes	Match	Last Edit
IPQC	-	No damage, moisture free, fungus free, Termite free, fitting properly with nails.	-	☐	-
IPQC	-	Imp - 0.2Amp, Bining color sticker on master box	-	☐	-
IPQC	-	No damage, Should be moisture free, proper printing if any	-	☐	-
IPQC	-	In every module	-	☐	-
IPQC	-	As per WI of Packing	-	☐	-
IPQC	-	Appropriate packing list with date and model number	-	☐	-
PDI	-	As per agreed visual criteria (AQL=2.5 for Major & AQL=4.0 for Minor)	-	☐	-
PDI	-	Length,width $\pm 1.5\text{mm}$ Dimension $\leq 1.5\text{m}$, Diagonal Difference $\leq 3\text{mm}$	-	☐	-
PDI	-	(The measured insulation resistance times the area of module shall not be $< 40 \text{ Mohms/mtr}^2$ @ system voltage) 1.5KV for 3 sec, $\geq 40 \text{ Mohms/mtr}^2$	-	☐	-
PDI	-	Test conditions: (System voltage X 2 + 1000) X 120%, Current Leakage is $\leq 50 \mu\text{A}$	-	☐	-
PDI	-	2.5 x of fuse rating for 3 sec, $\leq 0.1\Omega$	-	☐	-
PDI	-	Average power should be match with rated label Wp.	-	☐	-
PDI	-	As per agreed EL criteria (AQL=2.5 for Major & AQL=4.0 for Minor)	-	☐	-
PDI	-	1.5KV for 120sec, $\geq 40 \text{ Mohms} \cdot \text{mtr}^2$	-	☐	-
PDI	-	40N load applied, No mechanical damage, Power degradation not more than 5%, Insulation shall meet requirement.	-	☐	-

VISUAL / EL SPECIFICATIONS

Parameter	Customer Spec	Premier Spec	Remarks / Notes	Match	Last Edit
Cell	No visible cell crack is accepted	No visible cell crack is accepted	Any crack visible by naked eye without magnification or backlight	YES	-
Cell	-	Minor colour mismatch allowed from viewing Distance 1.0 meter. Major colour difference if effect on cell performance is not allowed.	Colour difference between cells	☐	-

Parameter	Customer Spec	Premier Spec	Remarks / Notes	Match	Last Edit
Cell	-	For the back of the bi-facial cells,color difference is accepted ,but yellow is unaccepted.	Colour difference between cells	☐	- -
Cell	-	Dust <3mm Ø accepted. Qty: 1no/cell, 3 no's in module	Cell should be free from the major dust particles.	☐	- -
Cell	-	< 3.0 mm Ø, Qty: 3 no's in module.	Cell should be free from the major dust particles in white area.	☐	- -
Cell	-	Minimum 0.5mm Light should be passing through gap.	The gap between cell to cell and string to string should be maintained	☐	- -
Cell	-	L≤ 5mm & not touching the grid Line. Q≤ 1 per cell, Q≤3 per module	Chip in cell	☐	- -
Cell	-	Not allowed	V chip or any other chip in cell extending to grid line	☐	- -
Cell	-	Minimum 0.3mm Light should be passing through gap. Cells short not allowed.	The gap between cell to cell and string to string should be maintained	☐	- -
Cell	-	≤2mm	Misalignment between Strings	☐	- -
Cell	-	Hair touching the active are - Not Allowed Hair in white area (Inactive area) - Allowed	Module should be free from the hair from active area.	☐	- -
Cell	-	Conductive flake L : ≥ 2mm Non-conductive flake L : > 15mm not accepted	Soldering balls and Backsheet flake on cell and inactive area	☐	- -
Cell	-	Back side: non-conductive foreign body L≤15mm, Q≤3 acceptable.	Module should be free from major foreign particles	☐	- -
Cell	-	Back side: The back of the cells string high temperature tape imprint and cells grid line diffusion is allowed.	Tape delamination is not allowed	☐	- -
Cell	-	Insects on the cell not accepted	Insects/ Flies in module will create bubbles.	☐	- -
Cell	-	Length of the scratch should be ≤50 mm and not visible from 1.0 meter distance, Qty 6 no's	Any scratch on cell visible by naked eye but not affecting the cell integrity or printing	☐	- -
Cell	-	Scratching on the back L≤50mm,Q≤8	-	☐	- -
Cell	-	The distance between cell and frame edge should not be less than 0.5 mm.	There is a minimum space between frame and cells with no risk of short circuit	☐	- -
Cell	-	≤ 2mm, Q≤8 cells in a module	Grid line missing in solar cells	☐	- -
Cell	-	Back finger grid line break point: L≤5mm,N≤3 / cell, Q≤10 / module. Finger was oxidized and became yellow, unacceptable.	Grid line missing in solar cells	☐	- -
Cell Conductors	-	Interconnect ribbon shall not deviate ≤2mm of width from the ideal position. Qty: 8 in a module. (If 60% ribbon is fully on busbar)	Misalignment of solar ribbon taperor parallel shifting ribbon twist, ribbon bend from front or back side of Cells.	☐	- -

Parameter	Customer Spec	Premier Spec	Remarks / Notes	Match	Last Edit
Cell Conductors	-	Interconnect ribbon shall not deviate ≤ 2 mm of width from the ideal position. Qty: 10 in a module. (If 60% ribbon is fully on busbar)	Misalignment of solar ribbon taper or parallel shifting ribbon twist, ribbon bend from front or back side of Cells.	☐	- -
Cell Conductors	-	≤ 2 mm allowed	Extra interconnect from busbar - it should be not create risk.	☐	- -
Cell Conductors	-	Gap should be minimum 1mm.	There is a minimum space between cell edge and busbar with no risk of short circuit	☐	- -
Glass	-	Scratch ≤ 10 mm is ignored Scratch ≤ 50 mm Qty 4 no's, no nail deep.	Glass should be free from any major scratch	☐	- -
Glass	-	Scratches on back side: ≤ 10 mm is ignored, ≤ 70 mm Qty 5 no's. not visible from 1.0 meter distance	Glass should be free from any major scratches	☐	- -
Glass	-	Difference ≤ 3 mm	No misalignment should be there between both glass.	☐	- -
Glass	-	L >20cm, Quantity 3 per module allowed	No sharp edges to be there on glasses	☐	- -
Glass	-	No Contamination on Glass >80mm not accepted (visible from 1.0 M)	Glass should be free from major contamination	☐	- -
Glass	-	No glass abrasion to be seen	Glass abrasion not allowed	☐	- -
Glass	-	A: Round bubble: 1)<0.3mm, ignored 2) $0.3 \leq D < 1.0$ mm=2 no's 3) $1.0 \leq D < 2$ mm=1no allowed 4)>2mm not allowed. B: Longitudinal bubbles: 1) L <0.5L ≤ 2.0 and W ≤ 1.5 =2 nos	The glass shall be free of any air bubbles and foreign inclusions	☐	- -
Encapsulation	-	Size ≤ 10 mm ² Q ≤ 6 no's No Bubbles Can Touch the Laminate Edge.	Any air bubbles visually apparent from the glass side of the EVA	☐	- -
Encapsulation	-	On Cells: ≤ 3 mm ² per module Q ≤ 6 no's	Any air bubbles visually apparent from the glass side of the EVA	☐	- -
Encapsulation	-	For Rear Side: Not allowed if Length of bubble is L>10 mm or Area > 4 mm ² Qty Allowed: 10no's within a Module other than cell area and not reducing creep age distance.	Any air bubbles visually apparent from the glass side of the EVA	☐	- -
Encapsulation	-	For Rear Side: Not allowed if Length of bubble is L> 15 mm or Area > 8 mm ² Qty Allowed:10 within a Module other than cell area and not reducing creepage distance	Any air bubbles visually apparent from the glass side of the EVA	☐	- -
Encapsulation	-	Non-conforming if length is L >10mm or A >20 mm ² , Qty Allowed:3 no's	The front cover/glass shall be free of EVA residue, especially in the active area	☐	- -
back sheet	-	Dents with D ≤ 0.5 mm or Length ≤ 50 mm , Q ≤ 3 per module	The backsheet shall be flat and should not contain any major dents.	☐	- -
back sheet	-	Scratch felt by fingers not allowed Scratch with any signs of surface peel off not allowed scratch not felt by finger, W ≤ 0.2 mm, or L	Any scratch (without nail feeling) in backsheet visible by naked eye	☐	- -

Parameter	Customer Spec	Premier Spec	Remarks / Notes	Match	Last Edit
		≤100mm and Q ≤3 , Back sheet cut not allowed			
back sheet	-	Not Allowed	Reverse glass & back sheet	☐	- -
bardode	-	Barcode shall be straight	Barcode should not be cross & not overlap on the cell.	☐	- -
Frame	-	The silicon overflow shall be smooth. Irregular silicon overflow of L ≤ 100mm, Qty ≤ 4 is accepted.	There should be no overflow of sealant on back- side	☐	- -
Frame	-	Sealant should not be protruding from the frame.	There should be no overflow of sealant on front side	☐	- -
Frame	-	No frame deformity is accepted,	The Frame should be free from deformity	☐	- -
Frame	-	Scratch L ≤ 100mm, Qty: 10 no's	The Frame should be free from the major scratches	☐	- -
Frame	-	Anodizing thickness ≥15 microns	It should be >15 microns	☐	- -
Frame	-	Deep anodization layer damages are not accepted.	The Frame should be free from the anodize coating layer damages.	☐	- -
Frame	-	Mismatch front & Back >0.5 mm not accepted	There should not be any miss-alignment between frames.	☐	- -
Frame	-	>0.5 mm (Front & Back) not accepted	There should not be any gap and miss-alignment between frames.	☐	- -
Frame	-	Sharp edges are not allowed	Edges of the frame should be processed in manner to Eliminate any risk of injury during handling.	☐	- -
Frame	-	>100mm silicone glue dirt not accepted	Frames should be clean with out any major marks & rust.	☐	- -
Frame	-	As per drawing	Grounding holes to be as per specification	☐	- -
Junction box	-	≥ 10 degree rotation or >5 mm tilt is not accepted	Junction should be properly aligned and placed	☐	- -
Junction box	-	Partially filled potting material is not allowed. (It should be fully covered)	Junction box fully filled with potting material	☐	- -
Junction box	-	Should be uniform	It should be uniform and no air gaps/ voids of RTV.	☐	- -
Junction box	-	No damage is accepted	No damage is accepted	☐	- -
Cell	-	Accepted if: Crack length ≤ 40mm ignored Crack length >40mm - <75mm Crack No's - N ≤ 8 per module. Full cell crack not acceptable.	Crack in solar cell creating dark area on either side isolating a particular area of cell.	☐	- -
Cell	-	"x" 10mmx10mm ignored, >10mm to ≤ 30mm, for MBB 5% of total cells "v" crack <10mm	"x" or "v" type	☐	- -

Parameter	Customer Spec	Premier Spec	Remarks / Notes	Match	Last Edit
		ignored, >10mm to ≤ 30mm, for MBB 15% of total cells.			
Cell	-	5 % of cell area, 7% of cells in module	Area of a grid finger with a lower luminescence. Interrupted or non-existent grid finger is Generated during the solar cell printing process	☐	-
Cell	-	Not Allowed	These are completely dark cells or shunted cells or the cells with the wrong polarity.	☐	-
Cell Conductors	-	Not Allowed	Interconnect/Ribbon not soldered with the cell -bus- bars.	☐	-
Cell Conductors	-	Poor soldering along with full length cell bus bar not acceptable. ≤15% area is allowed, 8 no's cells per module. Minor poor soldering area at edge of cell not counted.	The dark area in the cell is coming due to the improper soldering of the ribbon.	☐	-
Cell	-	if it is physically visible not allowed. Allowed if: ≤10% in a single cell & 4 No. of cell in a module	Dark area visible due to broken chips / cracks	☐	-
Cell Conductors	-	Interconnect ribbon shall not deviate ≤4mm of width from the ideal position. Qty: 8 in a module. (If 60% ribbon is fully on bus bar).	Interconnect is not soldered to cell and crossed from the cell pad.	☐	-
Cell	-	For Dark Area Allowed if: Area is ≤15% 5 per module. For Dark Spot Allowed if: Area is ≤10% 10 per module.	Cloudy area with a lower luminescence. They get created due to Firing process	☐	-
Cell	-	If the Gray Value difference is between 20 % to 60 % Quantity and in MBB 10% of total Cells. Reference gray scale mentioned below.	Difference in brightness in the EL image due to different cell efficiency or mixing of different cell makes	☐	-
Cell	-	Ignore if Length of scratch is < 20 mm If Length of scratch is between 20 mm to 80 mm Qty Allowed: 10 cells per Module.	There are non Hidden crack type black lines on the cell	☐	-
Cell	-	Not Allowed	Cracks which are in more than one direction or more than one crack connecting to each other.	☐	-

BOM SUMMARY

Vendor Lock-in: Premier Energies

RFID Location Lock-in: TOP Left Side, front view

Document Ref: BOM_CUS_PEI-144-XXX-THGB-M10 Rev.0, 09-12-2025: BOM-CUS-0001-20251209

Component	Selected Model	Sub-Vendor	Spec / Notes
Solar Cell	PEPPL_PIRANHA_P TYPE_M10_BI-FACIAL	Premier Energies	Mono Crystalline Bifacial Perc10bb 182 x 91± 1mm, 170 ± 17.5

Component	Selected Model	Sub-Vendor	Spec / Notes
Front Cover	(D) AR Coated Semi Tempered HS Solar Glass	Xinyi Solar (Malaysla) SDN. BHD Xinyi Solar (SuZhou) Ltd. Guangxi Xinyi Photovoltaic Industry Co.,Ltd. Xinyi PV Products (Anhui) Holdings Ltd	(D) Thickness [mm]: 2mm Surface treatment: AR Coated Transmission data: ≥ 91.5%
Back Cover	(D) Semi tempered HS with Grid Patterned Glass	Xinyi Solar (Malaysla) SDN. BHD Xinyi Solar (SuZhou) Ltd. Guangxi Xinyi Photovoltaic Industry Co.,Ltd. Xinyi PV Products (Anhui) Holdings Ltd	(D) Thickness [mm]: 2mm Surface treatment: AR Coated Transmission Data: ≥ 91.6%
Cell Connector	(6) dia 0.26+/-0.04mm	Valeo	Dimensions [mm]: Ø-0.32/0.30/0.28/0.26mm Ø:0.24+/-0.04mm Composition of alloy: Sn60Pb40
String Connector	(E) Silver/Black	Valeo	Dimensions [mm]: 4(+2)x0.4mm Composition of alloy:Sn60Pb40
Frame	D	YuanshuoMetal Technology Co., ltd.	—
Junction Box	DSJB12y (y=c)	Dhash PV Technologies pvt, Ltd	Max. voltage [V]: 1500 Max. current [A]: 30
MC4 Compatible Connector	DS01	Dhash PV Technologies pvt, Ltd	Max. voltage = 1500VDC Max. Current =40A Max Temp = 100°C
Sealant (Adhesives)	HT906Z	Shanghai Huitian New Material Co., Ltd.	One component Silicon sealant
Potting Material	SKF323-AB	Guangzhou Baiyun Chemical Industry Co. Ltd	RTV silicone potting compound
EVA (Encapsulation) Front	EP 304	Hangzhou First Applied Material Co., Ltd/First Material Science (Thailand) Co., Ltd/ Vietnam Advance Film Material Company Limited	Thickness[mm]: 0.50±0.05 CTI: 600V Max. Storage temp. [°C]: ≤ 30
EVA (Encapsulation) Back	CO-557	Changzhou Sveck PV New Material Co., Ltd	Thickness[mm]: 0.50±0.05 CTI: 600V Max. Storage temp. [°C]: ≤ 30
Bypass Diode	40SQ045	PanJit International Inc.	Tj max= 200 °C
Solar Cable	5) 62930 IEC 131 1x4mm ²	Apar Industries Ltd	5) Max. Voltage = DC 1.5KV Upper limit temp(°C):120
Fluxing Agent	CX-700	ZHUHAI CHANGXIAN NEW MATERIAL TECHNOLOGY CO., LTD.	Low Residue liquid soldering flux
Additional Material (Fixing Tape and Insulation Tape)	FF-3665 / 3M 'Anti-UV PET Tape UV-1	Cybrid / 3M Material Technology	Cybrid / 3M Material Technology
RFID Tag	PID-X40	Perfect ID	PID-X40 wet inlay

EVENTS / EDIT LOG

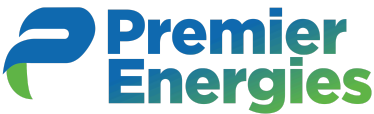
No edits recorded.

APPROVALS TRAIL

Level / Stage	Approver	Action	Comment / Notes	Timestamp
Level 2 ▪ Production	manoj	APPROVED	1 comment	09 Dec 2025, 13:33
Level 4 ▪ Technical Head	baskara	APPROVED	1 comment	09 Dec 2025, 13:33

COMMENTS

#	By	Timestamp	Comment
1	manoj	09 Dec 2025, 13:33	#1: test #2: test #3: test



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